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Lysosome pH and GSL Measurement in TMEM175 M393T Mutation Carriers

PPMI Project ID: 199

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Summary (or Abstract)

PPMI cell lines were differentiated into neural progenitor (NP) cells using the protocol noted below. The lysosome pH of these NP cells was measured using LysoSensor Yellow/Blue; no significant differences were observed between the lines. NP cells were also sent to NextCea for glycosphingolipid (GSL) measurements by LCMS. Lipid amounts were normalized to the total amount of protein in each sample to account for differences in input. There were no consistent differences observed between lines for any of the lipids measured (glucosylceramides, galactosylceramides, glucosylsphingosine, and glucosylcholesterol).

Method

Cell Culture and Differentiation

iPSCs were cultured on laminin 521 (Sundbyberg, Sweden) coated plates in mTeSR plus media. Cells were differentiated to neural progenitors as in Li and colleagues (2011). Prior to differentiation, cells were seeded onto plates coated with a combination of laminin 521 and 111. Differentiation was initiated by addition of DMEM/F12:Neurobasal Plus (1:1) media mixture containing 1X N2 supplement, 1X B27 supplement, 3 uM CHIR99021, 2 uM SB431542, 0.1 uM compound E, and 10 ng/mL leukemia inhibitory factor. Cells were maintained in this media for 7 days with daily media changes. After 7 days, cells were replated using Accutase onto Matrigel coated plates and cultured in the same media lacking compound E (replaced daily). Following a further 7 days of differentiation, cells were detached with Accutase, counted, and either plated for experiments or frozen in StemDiff Neural Progenitor freezing media for later use. Compound E, LIF, mTeSR plus, CHIR99021, SB431542, StemDiff freezing media, were from STEMCELL TECHNOLOGIES, (Vancouver, Canada). DMEM/F12, Neurobasal Plus, and N2/B27 supplements were from ThermoFisher. Matrigel was from Corning.

Lysosome pH Measurements

Neural progenitor cells derived from iPSCs were plated in Neurobasal Plus + B27 (Gibco) on Matrigel coated 96-well plates (Corning), and maintained at 37°C, 5% CO_2. On the day of the experiment, the media was aspirated from the cells and warm imaging buffer (mM: 135 NaCl, 5 KCL, 2 CaCl2, 1 MgCl2, 10 HEPES, 10 glucose, pH 7.4) containing 3 uM LysoSensor Yellow/Blue DND-160 (Invitrogen) was added. Cells were immediately imaged with a 20X objective on an IN Cell 6000 high content microscope using a 405 nm laser coupled with DAPI and FITC emission filters. The ratio of the DAPI:FITC emission (Blue/Yellow) was used to quantify pH in lysosomes defined by the Fast Puncta algorithm using In Carta software.

Lipid Measurements

Neural progenitor cells were thawed and washed with cold PBS and pelleted by centrifugation at 200xg for 5 minutes. Samples were sent to NextCea (Woburn, MA) for LCMS-based measurement of lipid species. Briefly, UPLC-MS/MS was used to quantitate molecular species of glucosylceramide (GluCer), galactosylceramide (GalCer), glucosylsphingosine (GluSph), and glucosylcholesterol (GluChol) on a SCIEX 7500 UPLC-MS/MS System. Different lipid classes were extracted using optimized liquid-liquid extraction procedures. Calibration curves were prepared from authentic standards using a class-based approach. An internal standard (i.e., structurally similar or stable-isotope labeled analog) was used for each analyte reported. Glucosylated and galactosylated forms of Cer/Sph were chromatographically separated and distinguished based on retention time. Resulting lipid amounts were normalized to total protein in each sample.

References

1. Li W, Sun W, Zhang Y, Wei W, Ambasudhan R, Xia P, Talantova M, Lin T, Kim J, Wang X, Kim WR, Lipton SA, Zhang K, and Ding S. (2011) Rapid Induction and long-term self-renewal of primitive neural precursors from human embryonic stem cells by small molecule inhibitors. *Proc Natl Acad Sci USA* 108(20):8299-8304.

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